

# Appendix A: Standard Operating Procedure - Benthic Harmful Cyanobacterial Bloom Sample Collection to Evaluate Water Quality to Inform Water Recreational Advisories

*Standalone copy of this document available for download at <https://mywaterquality.ca.gov/cyanoHab/docs/toxic-algal-mats-guidance-appendix-a.pdf>.*

## Scope

This document is prepared by the Benthic Guidance Subcommittee of the [CCHAB Network](#). This is a standard operating procedure (SOP) for collecting benthic (bottom) harmful cyanobacterial bloom (HCB) measurements and samples in the field and developed as part of the Benthic HCB Guidance. Monitoring HCBs using standardized processes is critical to evaluating water quality conditions that inform the potential exposure and health impacts associated with water recreation and inform decisions about posting advisories. Implementation of SOPs by trained personnel ensures that the resulting data are of known quality and comparable among monitoring entities. This SOP is designed to evaluate flowing channels (streams, rivers) and ponded waterbodies (lakes, reservoirs, etc.). Due to the constraints of this SOP's development, it is not intended for comprehensive ambient water quality monitoring or to inform impacts to aquatic life.

## Constraints

This SOP provides a rapid assessment of benthic HCBs to inform water recreation advisories, and as a conservative approach to protect public health, some steps will bias the analysis to a positive detection of benthic HCBs to lower the rate of false negatives. The design of this SOP was constrained by several elements highlighted by participants of the CCHAB Network and implementing entities of the first version of benthic HCB guidance published in 2020. As a result, the SOP is designed to meet low resource needs by requiring no scientific equipment (e.g., quadrats, placement of transects) and low-cost field supplies. In addition, the procedures were designed to be easily implemented by monitoring groups (i.e., health agencies, community groups and tribes) with web-based training. Cyanotoxins analysis is one of the possible indicators to trigger advisory postings, but efforts were made to include non-cyanotoxin indicators (e.g., visual assessment of cyanobacteria percent cover) for partners with low resources and to address the lack of governmental regulations for benthic HCBs. Thereby, the trigger levels for posting advisories for water recreation are based on a variety of quantitative indicators.

## Personnel Qualifications/Responsibilities

All personnel conducting this procedure must be familiar with the health and safety protocols within the [Health and Safety Guide](#) before visiting a site. Monitoring entities are responsible for their personnel implementing the procedures outlined in this SOP and to ensure that the data generated meets the standards and objectives of their monitoring project. The project or field crew leader should conduct periodic reviews of field personnel to ensure that personnel are following health and safety protocols and procedures in this SOP. For questions and to request training for this SOP, contact [CyanoHAB.Reports@waterboards.ca.gov](mailto:CyanoHAB.Reports@waterboards.ca.gov).

## Equipment and Supplies

When selecting equipment, consider project needs and select your equipment accordingly. Be sure that any materials that are used are in accordance with the lab's analytical method (refer to the last section for lab communication). The following is a general list of equipment for the field:

- |                                                                                                                    |                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Copy of this SOP                                                                          | <input type="checkbox"/> Measuring tape                                                                                                           |
| <input type="checkbox"/> Benthic Field Data Sheet ( <a href="#">Appendix C</a> )                                   | <input type="checkbox"/> Composite sample <sup>8</sup> container, amber or opaque glass <sup>9</sup> bottle with lid, minimum 60 mL <sup>10</sup> |
| <input type="checkbox"/> Chain of Custody (COC) Form                                                               | <input type="checkbox"/> Algal mat collection <a href="#">tools</a> such as scraper (spatula, putty knife), forceps, turkey baster (optional)     |
| <input type="checkbox"/> Waterproof labels for sample containers and permanent marker                              | <input type="checkbox"/> Bathyscope or <a href="#">DIY version</a> of underwater viewer (optional)                                                |
| <input type="checkbox"/> Clear tape to cover label (if not waterproof)                                             | <i>For cleaning tools (to decontaminate between waterbodies):</i>                                                                                 |
| <input type="checkbox"/> Resealable zipper plastic bags                                                            | <input type="checkbox"/> <i>Liquinox detergent (or other phosphate free detergent)</i>                                                            |
| <input type="checkbox"/> Gloves of appropriate length to prevent contact with the water                            | <input type="checkbox"/> <i>Wash basin</i>                                                                                                        |
| <input type="checkbox"/> Global Positioning System (GPS) receiver or smart phone capable of displaying coordinates | <input type="checkbox"/> <i>Soft brush or sponge</i>                                                                                              |
| <input type="checkbox"/> Pen and clipboard                                                                         | <input type="checkbox"/> <i>Cloth or paper to dry equipment</i>                                                                                   |
| <input type="checkbox"/> Cooler and ice gel packs                                                                  |                                                                                                                                                   |

## Procedures Overview

The procedures overview is repeated in [Attachment B](#) to provide a standalone handout. This procedure can be conducted on a flowing channel (e.g., streams, rivers) or along the beach of a ponded waterbody (e.g., lake, reservoir, pond). For clarity, this procedure often refers to a flowing channel since it is the most common type of California waterway sampled for benthic HCB using this approach. Key concepts of the procedure are listed as follows:

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<sup>8</sup> Do not add sodium thiosulfate to samples because it will degrade some classes of cyanotoxins.

<sup>9</sup> Plastic containers can adsorb toxins produced by cyanobacteria causing low quality data.

<sup>10</sup> Greater volume of sample may be required by laboratories to perform all requested types of analyses. Wide mouth containers are recommended.

- Sampling area – Size of sampling area is approximately 150 linear feet of beach or channel. Recreational areas should be targeted but if the substrate was previously disturbed by foot traffic, then move the sampling area upstream where potential benthic HCB growth is observed.
- Data collection and measurements - For a sampling area, personnel will record observations and visually estimate percent coverage of mats on a field data sheet. Visual hazards (e.g., detached and floating mats) associated with benthic HCBs and other associated data will also be recorded in the field data sheet.
- Number of samples collected - For a sampling area, personnel will collect 1 composite mat sample.

Monitoring entities have 3 procedure options to choose from based on waterbody types and resource constraints (Table 1). Options A and B are combinations of in-water and shoreline evaluations. Due to potential constraints of some monitoring entities in California, Option C does not require field personnel to not enter the water to accommodate lack of authorization or access to field supplies.

**Table 1. Field Procedure Options**

| <b>Field Procedure (pick one)</b>                                              | <b>Percent Coverage</b><br>(categories: 0%/not present, <15%, 15-50% and greater than 50%)                                                                                                                                                                                | <b>Composite mat sample</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. Combination of <i>In-water and shoreline</i> procedure for flowing channels | Percentage cover is visually estimated in areas with the most spatial coverage of mats observed along the shoreline and observed in-water. Hot spots are identified after surveying the full sample area. (Figure 3A, 3B)                                                 | Fill container $\frac{1}{4}$ full with ambient water. Collect mat samples at 4 hot spots, 2 hot spots observed along the shoreline and 2 hot spots observed in-water. Hot spots can include detached and floating mats found in-water or on the shoreline. At each hot spot collect 3 pieces (approx. 3 cm <sup>2</sup> ) of visually distinct mats to collect diverse material. (Figure 6A). Total of 12 pieces collected for the composite.                                                       |
| B. Combination of <i>In-water and shoreline</i> procedure for ponded waterbody | Percentage cover is visually estimated in areas with the most spatial coverage of mats observed along the shoreline and observed in-water. Hot spots are identified after surveying the full sample area. (Figure 3A, 3B) Limit depth of water to knee deep or 0.5 meter. | Fill container $\frac{1}{4}$ full with ambient water. Collect mat samples at 4 hot spots, 2 hot spots observed along the shoreline and 2 hot spots observed in-water. Hot spots can include detached and floating mats found in-water or on the shoreline. At each hot spot collect 3 pieces (approx. 3 cm <sup>2</sup> ) of visually distinct mats to collect diverse material. (Figure 6A). Total of 12 pieces collected for the composite.                                                       |
| C. <i>Shoreline only</i> procedure for both waterbody types <sup>11</sup>      | Percentage cover is visually estimated at areas with the most lateral growth observed along the shoreline and near shore area. Hot spots are identified after surveying the full sample area. (Figure 3C)                                                                 | Fill container $\frac{1}{4}$ full with ambient water. Collect mat samples at 2-4 hot spots along the shoreline. Hot spots can include detached and floating mats found in near shore area or on the shoreline. At each hot spot collect 3 pieces (approx. 3 cm <sup>2</sup> ) of visually distinct mats to collect diverse material. (Figure 6B). If there are fewer than 4 hot spots, collect additional pieces at hot spots identified to equal a total of 12 pieces collected for the composite. |

<sup>11</sup> Due to constraints of monitoring entities in California, this provides an option for field personnel to not enter the water due to lack of authorization or field supplies. If this is not a constraint for your monitoring entity, implement in-water Options A or B.

## Procedure - Field Survey

*This section provides the steps to prepare for sample collection. For additional information on field preparation, refer to the following standard operating procedures: [Site Reconnaissance Guide](#); [Collection of Field Data for Bioassessment](#).*

1. Prepare for sample transport
  - 1.1. Contact the laboratory prior to shipping samples and coordinate shipping timeframes, shipping conditions, and accompanying paperwork (i.e., COC form). Also check with mail carrier to determine shipping deadlines for overnight shipping to the laboratory.
2. Site selection considerations
  - 2.1. During implementation of this SOP, monitoring entities can make minor modifications to the sampling area due to the unique characteristics of the recreational areas of interest. Please note any modifications in the additional notes section of the field data sheet. With this flexibility, CCHAB Network appreciates detailed attention to all elements of this SOP to ensure evaluations are based on standardized processes.
  - 2.2. Sampling area selection should target recreational areas where exposure would be expected. Some site-specific considerations support moving the sampling area while minimizing the distance from recreational areas. Site-specific considerations include a) avoiding heavily disturbed areas like portions of beaches with heavy foot traffic and boat launches, and b) if hot spots are identified during the site reconnaissance, site selection can be moved to capture the worst-case exposure scenario.
  - 2.3. Riffles and non-stagnant water are suitable for colonization of toxic cyanobacteria like *Microcoleus sp.* so targeting these areas is recommended.
  - 2.4. Benthic HCBs can grow year-round in California depending on environmental conditions so this SOP can be implemented during any season.
3. Prepare for sample collection
  - 3.1. Set up sampling equipment and supplies in the staging area near the sample site.
  - 3.2. Complete labels with the following information:

This SOP commonly refers to hot spots and visual hazards of benthic HCBs.

*Hot spots* are areas of higher benthic growth of potential cyanobacteria and representative of the worst-case exposure scenario.

*Visual hazards of benthic HCBs* pose increased risk of exposure to the public, particularly children and dogs. These hazards include a.

Cyanobacteria mats (detached or growing) on the shore, b. Detaching/ breaking apart cyanobacteria mats in sampling area, and c. Floating cyanobacteria mats.

- 3.2.1. Station Code -or- Site Name (Waterbody name and landmark);  
Date; Sample Type = benthic composite.
- 3.3. If using a waterproof pen, pre-label containers at each site. If a waterproof pen is not supplied, complete labels at each site and apply them to the container after sampling. Cover label with clear tape to prevent loss of text.
4. Field data sheet
  - 4.1. Record applicable site data on the top area of the **Field Data Sheet** (Figure 1).
  - 4.2. Determine the coordinates of the sampling area by using a GPS receiver or smartphone. The northernmost or upstream point should represent the location.

| California FHAB Program                                               |  | Benthic Cyanobacteria Field Data Sheet                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | Version 2024.09.06                                                                                                                                    |  |
|-----------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Waterbody and site name:                                              |  | Date (mm/dd/yyyy):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | Personnel:                                                                                                                                            |  |
| Coordinates (dec. deg.):                                              |  | Sample Time (24:00):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | Agency:                                                                                                                                               |  |
| What type of field visit is being conducted? (check one)              |  | <input type="checkbox"/> <b>Event response.</b> Field visit initiated by observation or bloom report; no prior sampling or sampling frequency > 2 weeks.<br><input type="checkbox"/> <b>Follow-up monitoring.</b> Field visit initiated by previous event response; monitoring frequency is ≤ 2 weeks.<br><input type="checkbox"/> <b>Routine monitoring.</b> Field visits occur during pre-determined timeframe; monitoring occurs every 1-2 weeks.<br><input type="checkbox"/> <b>Other (please describe):</b> |  |                                                                                                                                                       |  |
| (1) How is cyanobacteria coverage assessed in 150 ft sampling area?   |  | <input type="checkbox"/> <b>Shoreline estimate.</b> Cyanobacteria percent cover was estimated based on observations from shore.<br><input type="checkbox"/> <b>In-water estimate.</b> Percent cover was estimated by wading in zig-zag patterns within waterbody.<br><input type="checkbox"/> <b>Other (please describe):</b>                                                                                                                                                                                    |  | Percent cover diagram:                                                                                                                                |  |
| (2) Are cyanobacteria present?                                        |  | <input type="checkbox"/> <b>Yes.</b> Take photos of cyanobacteria. What genera are present (if known)?<br><input type="checkbox"/> <b>No.</b> Document other algae below.                                                                                                                                                                                                                                                                                                                                        |  | <input type="checkbox"/> Anabaena<br><input type="checkbox"/> Oscillatoria<br><input type="checkbox"/> Microcoleus<br><input type="checkbox"/> Other: |  |
| (3) Where or how are the benthic mats growing (check all that apply)? |  | <input type="checkbox"/> in riffle <input type="checkbox"/> near shore <input type="checkbox"/> in pool or run <input type="checkbox"/> on roots, plants, or algae<br><input type="checkbox"/> on rocks <input type="checkbox"/> on sediment <input type="checkbox"/> in backwater area <input type="checkbox"/> in spires or bubble towers                                                                                                                                                                      |  |                                                                                                                                                       |  |
| (4A) Check any of the visual hazards that apply.                      |  | <input type="checkbox"/> Cyanobacterial mats are growing in puddles on shore, or have detached and accumulated onshore.<br><input type="checkbox"/> Mats are peeling from substrates, breaking apart, or overlapping, i.e., there is risk for detachment.<br><input type="checkbox"/> Floating mats observed in stagnant areas or flowing downstream during survey.                                                                                                                                              |  |                                                                                                                                                       |  |
| (4B) If present, how are benthic cyanobacterial mats distributed?     |  | <input type="checkbox"/> <b>Dispersed throughout reach.</b> Cyanobacteria are growing throughout the waterbody.<br><input type="checkbox"/> <b>Concentrated.</b> Cyanobacteria are localized in specific areas. How many areas? _____<br>In addition to above, check here if waterbody is too wide (>30 ft) for shoreline visuals or wading.                                                                                                                                                                     |  |                                                                                                                                                       |  |
| (4C) What is size class of largest mat patch?                         |  | <input type="checkbox"/> ≤ basketball <input type="checkbox"/> ≤ doormat <input type="checkbox"/> ≤ sedan <input type="checkbox"/> > sedan <input type="checkbox"/> Other:                                                                                                                                                                                                                                                                                                                                       |  |                                                                                                                                                       |  |
| (4D) Using diagram, what is total estimated % cover?                  |  | <input type="checkbox"/> not present <input type="checkbox"/> <15% cover <input type="checkbox"/> 15-50% cover <input type="checkbox"/> >50% cover                                                                                                                                                                                                                                                                                                                                                               |  |                                                                                                                                                       |  |
| (5) Are benthic mats being collected for lab analysis?                |  | <input type="checkbox"/> <b>Yes.</b> Are SPATTs being deployed? <input type="checkbox"/> <b>Yes.</b> Every _____ wk.<br><input type="checkbox"/> <b>No.</b>                                                                                                                                                                                                                                                                                                                                                      |  | Document SPATT location (if apply):                                                                                                                   |  |
| (6) If collecting benthic mats, how is the sample collected?          |  | <input type="checkbox"/> <b>From the shore.</b> 2 shore hotspots x 3 pieces of mats from each hotspot (+ 3 pieces from floating mats, if present)<br><input type="checkbox"/> <b>From shore and water.</b> 2 shore hotspots + 2 water hotspots x 3 pieces each (+ 3 pieces from floating mats, if present)                                                                                                                                                                                                       |  |                                                                                                                                                       |  |
| (7) How will samples be analyzed in the lab (check all apply)?        |  | <input type="checkbox"/> Anatoxins <input type="checkbox"/> Microcystins <input type="checkbox"/> Saxitoxins <input type="checkbox"/> Cyanobacteria ID (do not freeze)<br><input type="checkbox"/> Cylindrospermopsins <input type="checkbox"/> Nodularins <input type="checkbox"/> qPCR (do not freeze) <input type="checkbox"/> Other:                                                                                                                                                                         |  |                                                                                                                                                       |  |
| Other field data (if collected)                                       |  | Water temp (°C):    Sp. cond (uS/cm):    Salinity (ppt):    Diss. O2 (%):    Diss. O2 (mg/L):    pH:    pH (mV):    Turbid (NTU):    Bar. pressure:                                                                                                                                                                                                                                                                                                                                                              |  |                                                                                                                                                       |  |
| Additional notes or observations:                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                       |  |

Figure 1. Top area of the field data sheet (shown in red square) for recording waterbody information, date and time, sampler information, and monitoring type.

## Coverage Assessment

This section provides the steps for visually assessing benthic cyanobacteria, including out of water (shoreline survey) and in water (wadeable vs. non-wadeable channels; ponded waterbodies) surveys. Methods closely follow the accompanying Field Data Sheet, which is designed to walk personnel through the collection process. For additional information on field surveys, refer to the following standard operating procedures: [Site Reconnaissance Guide](#); [Collection of Field Data for Bioassessment](#).

5. Coverage Assessment
  - 5.1. Survey of sampling area

- 5.1.1. The survey of the sampling area is limited to a maximum of 150 ft of beach or channel reach. If the area of interest exceeds this length, conduct a second assessment in a separate area.
- 5.1.2. Use measuring tape to mark approximately 150 feet using stakes, coolers, or other objects to keep track of the linear sampling area.
- 5.1.3. *Visual reconnaissance*. Walk along the entire reach of the sampling area while looking for algal and cyanobacteria mats on the shore, floating on the water surface, and submerged below the water surface. Differentiating between submerged algae and cyanobacteria from the shore is challenging. Therefore, for the purposes of this protocol and a conservative approach to inform advisories, when water clarity is low (especially in faster moving water), assume submerged materials are cyanobacterial mats<sup>12</sup> and note the turbid conditions on the field data sheet. A bathyscope or homemade underwater viewer may make it easier to distinguish between submerged material and cyanobacterial material.
- 5.1.4. Refer to section 5.2 for detailed instructions to perform each procedure, option A, B, or C.
- 5.1.5. Record how benthic cyanobacteria coverage is assessed on the **Field Data Sheet, Section 1 (i.e., visual shoreline vs. in-water)**. Indicate presence and genera (if known) in **Section 2**, then record locations of benthic mat growth<sup>13</sup> in **Section 3** (Figure 2).

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<sup>12</sup> This assumption biases the decision to detecting cyanobacteria mats when what clarity is low. Positive bias is acceptable for public health decisions to be conservative.

<sup>13</sup> Definitions of areas that benthic growth can occur. Riffle - rocky part of channel with rough, cascading, or fast-moving water. Rocks - hard substrates including large gravels, cobbles, and boulders. Near shore - typically shallow area of water that is closest to shoreline. Sediment - finer substrate that is moved and deposited. Pool or run - deeper, slower moving area of a channel. Backwater - branched or parallel channel with little or no current. Roots, plants, or algae – materials that benthic HCBs can adhere while growing. Spires or bubble towers – consists of benthic cyanobacteria growth, see visual guide.



| California FHAB Program                                               |  | Benthic Cyanobacteria Field Data Sheet                                                                                                                                                                                                                                                                                                                                                                       |  | Version 2024.09.06                                |  |
|-----------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------|--|
| Waterbody and site name:                                              |  | Date (mm/dd/yyyy):                                                                                                                                                                                                                                                                                                                                                                                           |  | Personnel:                                        |  |
| Coordinates (dec. deg.):                                              |  | Sample Time (24:00):                                                                                                                                                                                                                                                                                                                                                                                         |  | Agency:                                           |  |
| What type of field visit is being conducted? (check one)              |  | <b>Event response.</b> Field visit initiated by observation or bloom report; no prior sampling or sampling frequency > 2 weeks.<br><b>Follow-up monitoring.</b> Field visit initiated by previous event response; monitoring frequency is ≤ 2 weeks.<br><b>Routine monitoring.</b> Field visits occur during pre-determined timeframe; monitoring occurs every 1-2 weeks.<br><b>Other (please describe):</b> |  |                                                   |  |
| (1) How is cyanobacteria coverage assessed in 150 ft sampling area?   |  | <b>Shoreline estimate.</b> Cyanobacteria percent cover was estimated based on observations from shore.<br><b>In-water estimate.</b> Percent cover was estimated by wading in zig-zag patterns within waterbody.<br><b>Other (please describe):</b>                                                                                                                                                           |  | Percent cover diagram:                            |  |
| (2) Are cyanobacteria present?                                        |  | <b>Yes.</b> Take photos of cyanobacteria. What genera are present (if known)?<br><b>No.</b> Document other algae below.                                                                                                                                                                                                                                                                                      |  | Anabaena<br>Microcoleus<br>Oscillatoria<br>Other: |  |
| (3) Where or how are the benthic mats growing (check all that apply)? |  | <input type="checkbox"/> in riffle <input type="checkbox"/> near shore <input type="checkbox"/> in pool or run <input type="checkbox"/> on roots, plants, or algae<br><input type="checkbox"/> on rocks <input type="checkbox"/> on sediment <input type="checkbox"/> in backwater area <input type="checkbox"/> in spires or bubble towers                                                                  |  |                                                   |  |
| (4A) Check any of the visual hazards that apply.                      |  | <input type="checkbox"/> Cyanobacterial mats are growing in puddles on shore, or have detached and accumulated onshore.<br><input type="checkbox"/> Mats are peeling from substrates, breaking apart, or overlapping. I.e., there is risk for detachment.<br><input type="checkbox"/> Floating mats observed in stagnant areas or flowing downstream during survey.                                          |  |                                                   |  |
| (4B) If present, how are benthic cyanobacterial mats distributed?     |  | <b>Dispersed throughout reach.</b> Cyanobacteria are growing throughout the waterbody.<br><b>Concentrated.</b> Cyanobacteria are localized in specific areas. How many areas?<br>In addition to above, check here if waterbody is too wide (>30 ft) for shoreline visuals or wading                                                                                                                          |  |                                                   |  |
| (4C) What is size class of largest mat patch?                         |  | <input type="checkbox"/> ≤ basketball <input type="checkbox"/> ≤ doormat <input type="checkbox"/> ≤ sedan <input type="checkbox"/> > sedan    Other:                                                                                                                                                                                                                                                         |  |                                                   |  |
| (4D) Using diagram, what is total estimated % cover?                  |  | <input type="checkbox"/> not present <input type="checkbox"/> <15% cover <input type="checkbox"/> 15-50% cover <input type="checkbox"/> >50% cover                                                                                                                                                                                                                                                           |  |                                                   |  |
| (5) Are benthic mats being collected for lab analysis?                |  | <b>Yes.</b> Are SPATs being deployed? <b>Yes.</b> Every ____ wk.                                                                                                                                                                                                                                                                                                                                             |  | Document SPATT location (if apply):               |  |
| (6) If collecting benthic mats, how is the sample collected?          |  | <b>From the shore.</b> 2 shore hotspots x 3 pieces of mats from each hotspot (+ 3 pieces from floating mats, if present)<br><b>From shore and water.</b> 2 shore hotspots + 2 water hotspots x 3 pieces each (+ 3 pieces from floating mats, if present)                                                                                                                                                     |  |                                                   |  |
| (7) How will samples be analyzed in the lab (check all apply)?        |  | <input type="checkbox"/> Anatoxins <input type="checkbox"/> Microcystins <input type="checkbox"/> Saxitoxins <input type="checkbox"/> Cyanobacteria ID (do not freeze)<br><input type="checkbox"/> Cylindrospermopsins <input type="checkbox"/> Nodularins <input type="checkbox"/> qPCR (do not freeze)    Other:                                                                                           |  |                                                   |  |
| Other field data (if collected)                                       |  | Water temp (°C):    Sp. cond (µS/cm):    Salinity (ppt):    Diss. O2 (%):    Diss. O2 (mg/L):    pH:    pH (mV):    Turbid (NTU):    Bar. pressure:                                                                                                                                                                                                                                                          |  |                                                   |  |
| Additional notes or observations:                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                              |  |                                                   |  |

Figure 2. Area of the field data sheet (shown in red square) for recording the type of survey being conducted, whether cyanobacteria are present, and location of cyanobacteria.

## 5.2. Detailed instructions for each procedure Option A, B, or C

### 5.2.1. Option A. Wadeable Channel survey (in-water). It is recommended for a team of at least 2 personnel to conduct field assessments.

5.2.1.1. If 2 personnel are surveying the channel, begin downstream and wade 150 feet upstream in a zig-zag pattern, each person taking the left or right side and surveying from shoreline to mid-width (Figure 3A).

5.2.1.2. If 1 personnel are surveying the channel, wade 150 feet upstream in a zig-zag pattern to cover the width of the channel (Figure 3B).

5.2.1.3. Cyanobacteria species of concern can grow in stagnant or pooled areas, in slow moving water, and fast-moving areas (e.g., riffles). They can also be floating or stranded on the shoreline. Note where cyanobacteria growth is most dense and/or covers the most spatial area, these are the hotspots.

5.2.1.4. **Modification for wide or non-wadeable channel.** For wide channels or water currents that don't support safe wading, the width of the wading segment should extend a maximum of 30 feet towards the opposite bank or until a depth of knee deep (~ 0.5 m).

### 5.2.2. Option B. Ponded waterway survey (in-water).

5.2.2.1. If 2 personnel are surveying a ponded waterway, each personnel start at opposing ends of 150 feet area and wade towards each other in a zig-zag pattern from shore to deeper depth. The width of the wading segment should extend a maximum of 30 feet into the deeper water -or- until a depth of knee deep (~ 0.5 m).



- 5.2.2.2. If 1 personnel are surveying the ponded area, wade in a zig-zag pattern from the shore to deeper depth (up to knee deep or ~0.5 m) and back to the shore to cover more area.
- 5.2.2.3. Cyanobacteria species of concern can grow in stagnant or pooled areas, in slow moving water, and also fast-moving areas (e.g., riffles). They can also be floating or stranded on the shoreline. Note where cyanobacteria growth is most dense and/or covers the most spatial area, these are the hotspots.

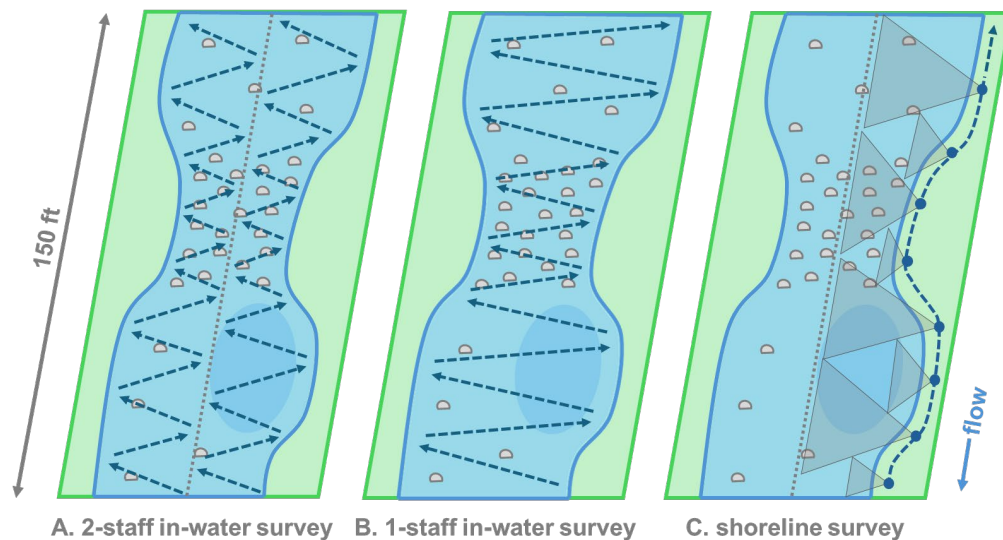


Figure 3. Schematic for conducting: A) in-water survey with 2-personnel wading in zig-zag patterns; B) in-water survey with 1-personnel wading in a zig-zag pattern; and C) shoreline survey with periodic stops for assessing nearshore and at least 30 ft offshore.

- 5.2.3. Option C. Shoreline Survey (out of water evaluation).
  - 5.2.3.1. For surveying a channel, begin at the downstream point of the sampling area and walk 150 feet upstream along the shoreline (Figure 3C). If surveying a ponded waterbody, walk towards the current windward side, if no wind is felt then walk approx. west to east.
  - 5.2.3.2. Cyanobacteria species of concern can grow in stagnant or pooled areas, in slow moving water, and fast-moving areas (e.g., riffles). Note where potential cyanobacteria growth is most dense and/or covers the most spatial area near or on the shoreline; these are the hotspots.
- 5.3. Estimating spatial coverage (Figure 4).
  - 5.3.1. Channel Estimations (in-water).
    - 5.3.1.1. If potential cyanobacterial mats are present in the sampling area, consider the total spatial coverage and growth patterns (density, patchiness) of this growth along the shore and across to the opposite side of the channel (if the channel is too deep to safely cross, note

the approx. distance of mid-channel width and walk back towards the shore).

- 5.3.1.2. To visually determine patterns across the entire sample area, it is helpful to divide the area into thirds or approx. 50 ft. portions (Figure 4). Pay attention to the spatial coverage and growth patterns for each of the three portions, then the portion with the greatest spatial coverage and unique growth patterns will be used to record final values in the field data sheet.

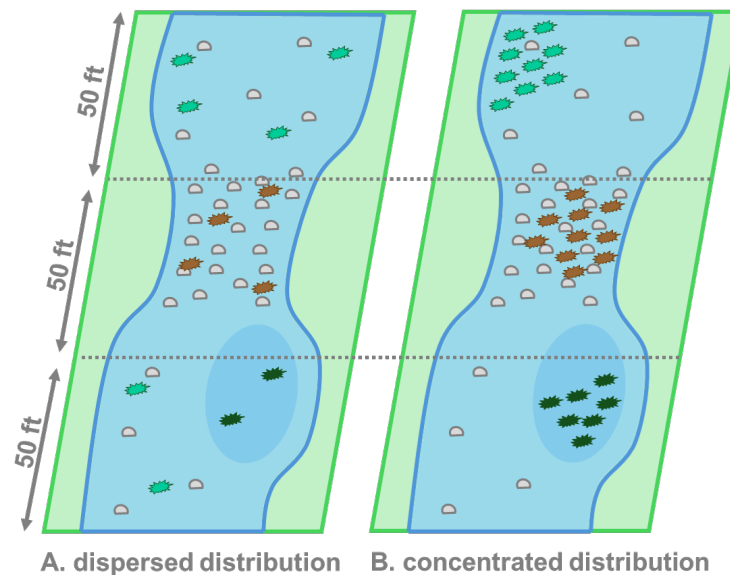


Figure 4. Schematic showing: A) dispersed distribution of benthic cyanobacteria depicted as light green, brown, and dark green morphotypes; and B) concentrated distribution. The survey reach can be broken into thirds or 50 feet segments to support visual coverage estimates.

- 5.3.1.2.1. Note: Some areas of growth may appear to accumulate more dense *vertical* growth, sometimes appearing as spires floating in the water column or growth of overlapping multiple types of algae and cyanobacteria. Density of growth can be noted, but spatial coverage is based on the *lateral* extent of potential cyanobacteria mats (growing attached, detached, and/or floating) across the specified area.
- 5.3.2. Shoreline Estimations (out of water evaluation).
- 5.3.2.1. If potential<sup>14</sup> cyanobacterial mats are present in the sampling area, consider the total spatial coverage of this growth along the shore and across the visible width of the channel (< 30 ft across). When more than 1 field personnel are available, the sampling area can be divided in half.

<sup>14</sup> This procedure often refers to “potential” cyanobacteria mats because identification is challenging for novice personnel. Web-based training and the visual guide provides examples of cyanobacteria morphology. A conservative approach incorporates potential cyanobacteria mats in visual estimates for percentage coverage and sample collection.

- 5.3.3. Record observations from either procedure option on the **Field Data Sheet, Section 4**, corresponding with parts A-D, the last step is recording the estimated percent cover.
- 5.3.4. *Part A. Determine if visual hazards<sup>15</sup> are present*; refer to Table 2 below and mark choices on **Section 4A of the Field Data Sheet** (Figure 5).

**Table 2. Visual Hazards of Cyanobacteria Mats**

| Benthic HCB visual hazard type                                                             | Description                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyanobacteria mats (detached or growing) on the shore                                      | Cyanobacteria mats can grow on the shore in shallow puddles and can become more easily detached from flow changes or water current. Mats that have become detached from the bottom substrate can also accumulate on the shore (dry mats remain hazardous to people and pets). |
| Detaching/breaking apart cyanobacteria mats in sampling area (still attached to substrate) | Mats that appear to be breaking apart, peeling from substrates, or overlapping can increase the risk of detaching. Mats that are densely growing (appearing concentrated) can become detached more easily, also increasing risk of exposure.                                  |
| Floating cyanobacteria mats                                                                | Floating mats can be observed in stagnant areas (e.g., swimming holes) or flow downstream when surveying.                                                                                                                                                                     |

California FHAB Program Benthic Cyanobacteria Field Data Sheet Version 2024.09.06

Waterbody and site name: \_\_\_\_\_ Date (mm/dd/yyyy): \_\_\_\_\_ Personnel: \_\_\_\_\_  
 Coordinates (dec. deg.): \_\_\_\_\_ Sample Time (24:00): \_\_\_\_\_ Agency: \_\_\_\_\_

What type of field visit is being conducted? (check one)  
☐ Event response. Field visit initiated by observation or bloom report; no prior sampling or sampling frequency > 2 weeks.  
☐ Follow-up monitoring. Field visit initiated by previous event response; monitoring frequency is < 2 weeks.  
☐ Routine monitoring. Field visits occur during pre-determined timeframe; monitoring occurs every 1-2 weeks.  
 Other (please describe): \_\_\_\_\_

(1) How is cyanobacteria coverage assessed in 150 ft sampling area?  
☐ Shoreline estimate. Cyanobacteria percent cover was estimated based on observations from shore.  
☐ In-water estimate. Percent cover was estimated by wading in zig-zag patterns within waterbody.  
 Other (please describe): \_\_\_\_\_

(2) Are cyanobacteria present?  
☐ Yes. Take photos of cyanobacteria. What genera are present (if known)? Anabaena \_\_\_\_\_ Oscillatoria \_\_\_\_\_  
☐ No. Document other algae below. Microcoleus \_\_\_\_\_ Other: \_\_\_\_\_

(3) Where or how are the benthic mats growing (check all that apply)?  
☐ In riffle ☐ near shore ☐ in pool or run ☐ on roots, plants, or algae  
☐ on rocks ☐ on sediment ☐ in backwater area ☐ in spires or bubble towers

(4) Check any of the visual hazards that apply.  
☐ Cyanobacterial mats are growing in puddles on shore, or have detached and accumulated onshore.  
☐ Mats are peeling from substrates, breaking apart, or overlapping. I.e., there is risk for detachment.  
☐ Floating mats observed in stagnant areas or flowing downstream during survey.

(4B) If present, how are benthic cyanobacterial mats distributed?  
☐ Dispersed throughout reach. Cyanobacteria are growing throughout the waterbody.  
☐ Concentrated. Cyanobacteria are localized in specific areas. How many areas? \_\_\_\_\_  
 In addition to above, check here if waterbody is too wide (>30 ft) for shoreline visuals or wading

(4C) What is size class of largest mat patch?  
☐ < basketball ☐ < doormat ☐ < sedan ☐ > sedan ☐ Other: \_\_\_\_\_

(4D) Using diagram, what is total estimated % cover?  
☐ not present ☐ <15% cover ☐ 15-50% cover ☐ >50% cover

(5) Are benthic mats being collected for lab analysis?  
☐ Yes. Are SPATs being deployed? ☐ Yes. Every \_\_\_\_\_ wk. Document SPAT location (if apply): \_\_\_\_\_  
☐ No. \_\_\_\_\_

(6) If collecting benthic mats, how is the sample collected?  
☐ From the shore. 2 shore hotspots x 3 pieces of mats from each hotspot (+ 3 pieces from floating mats, if present)  
☐ From shore and water. 2 shore hotspots + 2 water hotspots x 3 pieces each (+ 3 pieces from floating mats, if present)

(7) How will samples be analyzed in the lab (check all apply)?  
☐ Anatoxins ☐ Microcystins ☐ Saxitoxins ☐ Cyanobacteria ID (do not freeze)  
☐ Cylindrospermopsins ☐ Nodularins ☐ qPCR (do not freeze) ☐ Other: \_\_\_\_\_

Other field data (if collected)  
 Water temp (°C): \_\_\_\_\_ Sp. cond (µS/cm): \_\_\_\_\_ Salinity (ppt): \_\_\_\_\_ Diss. O2 (%): \_\_\_\_\_ Diss. O2 (mg/L): \_\_\_\_\_ pH: \_\_\_\_\_ pH (mV): \_\_\_\_\_ Turbid (NTU): \_\_\_\_\_ Bar. pressure: \_\_\_\_\_

Additional notes or observations: \_\_\_\_\_

<sup>15</sup> Visual hazards of HCBs pose increased risk of exposure to the public, particularly children and dogs. These are listed on the Field Data Sheet and included in Table 2.

*Figure 5. Area of the field data sheet (shown in red square) for recording information on hazards and the size and distribution of benthic blooms. Refer to the percentage cover diagram for estimations.*

- 5.3.5. *Part B. Identify the pattern of spatial coverage in the sampling area; and mark choices on **Section 4B of the Field Data Sheet** (Figure 5).*
  - 5.3.5.1. “Dispersed” throughout the area - cyanobacteria growth is observed throughout the sampling area with a range of densities (Figure 4A).
  - 5.3.5.2. “Concentrated” in patches - cyanobacteria growth appears patchy, concentrating in a few areas and not dispersed throughout the sampling area (Figure 4B). Patches can be connected to each other and dissipate among the connections of the patches; this is still considered “concentrated” growth.
- 5.3.6. *Part C. Identify the size class of cyanobacteria growth; and mark choices on **Section 4C of the Field Data Sheet** (Figure 5).*
  - 5.3.6.1. Estimate the size of the largest cyanobacterial mat growth in the assessment area, using prescribed categories on the Field Data Sheet
    - 5.3.6.1.1. basketball (smallest),
    - 5.3.6.1.2. doormat, -or-
    - 5.3.6.1.3. sedan (largest).
  - 5.3.6.2. Often when growth is patchy, there are many smaller patches and a single larger patch. In this case, check the box associated with the largest size class and record your observations in the notes section.
- 5.3.7. *D. Determine the percentage spatial coverage; and mark choices on **Section 4D of the Field Data Sheet** (Figure 5).*
  - 5.3.7.1. Refer to Figure 9 (Attachment A) to review a diagram for estimating percent cover categories. The diagram is provided for educational purposes and presents additional categories of percent coverage ranging from 2-75% to guide visual estimates.

When evaluating the approx. 150 ft long sampling area in thirds, field personnel will bias percent coverage based on the 50 ft reach with the greatest coverage. This approach will bias coverage estimates to be most conservative of protecting public health.
  - 5.3.7.2. Then consider your observations and pick the category that best fits your observation of the sampling area on the Field Data Sheet. Make one choice:
    - 5.3.7.2.1. 0% / not present,
    - 5.3.7.2.2. <15%,
    - 5.3.7.2.3. 15-50%, -or-
    - 5.3.7.2.4. > than 50%.

- 5.3.7.3. Tip: Divide the linear length of the sampling area into thirds, estimate the spatial coverage of each portion, then pick the greatest coverage category to represent the entire sampling area (Figure 4). This process biases decisions on percentage coverage to a greater value.
- 5.3.7.4. If no cyanobacterial mats are visible in the sampling area, note your observations and record remaining measurements on the field data sheet. Skip the next steps in this field protocol for sample collection.

## Cyanobacteria Mat Collection

*This section provides the steps for collecting a composite sample of benthic cyanobacterial mats for both out of water and in-water options. Composite sampling focuses on benthic cyanobacteria hotspots and hazards. For additional information on composite sampling, refer to the following standard operating procedures: [Collection of Field Data for Bioassessment](#).*

- 6. Options A and B. Combination of *in-water and shoreline* procedure for mat collection (For shoreline only procedure, go to step 7)
  - 6.1. *Composite cyanobacteria mat sample at shoreline and in-water, fill one sample container.* Fill the sample container approx.  $\frac{1}{4}$  of total volume with ambient water to provide moisture to collected material. Walk back along the surveyed area and collect the sample at 2 hotspots along the shoreline AND 2 hotspots in the water (Figure 6A). Hotspots are locations that you observe to have the densest or most area of coverage of cyanobacteria growth.
  - 6.2. At each hotspot, collect 3 pieces of mat material (each approx. 3 cm<sup>2</sup>) by pinching the material with gloved hands and dropping it into the same sample container. All 12 pieces of mat material are composited into the same sample container. Target mats with different appearances to produce a composite sample of diverse genera.
    - 6.2.1. Collection of a diverse population of mat materials is the goal of this composite sample. Target sample collection of visually distinct mats, to avoid collecting pieces of the same type of mat. Many species of cyanobacteria have different shades of color, texture, and density. If mat material appears visually uniform, collect pinches of material that are not side-by-side or connected to each other.
    - 6.2.2. Also, target mat material should include visual hazards like floating mats or those accumulated along the shoreline, as well as mats that are attached to the bottom substrate (cobble, sand). Faster moving water commonly supports the growth of HCB species that produce higher toxins. If the faster moving water area has potential cyanobacteria, then target this area for sample collection.
  - 6.3. If mat material cannot be pinched to remove it, use a scraper, forceps, turkey baster (when material is submerged), or other tool to grab approx. 3 cm<sup>2</sup> (approx. size of a quarter) of material to replicate a “pinch.” If collection of mat materials is

challenging, the sampler can use a bucket or bin to collect larger pieces of mat materials and then take the bin to a staging area to retrieve 3 pinched pieces of mat material for the sample container. Avoid collecting detritus, sand, pebbles, or invertebrates with your sample.

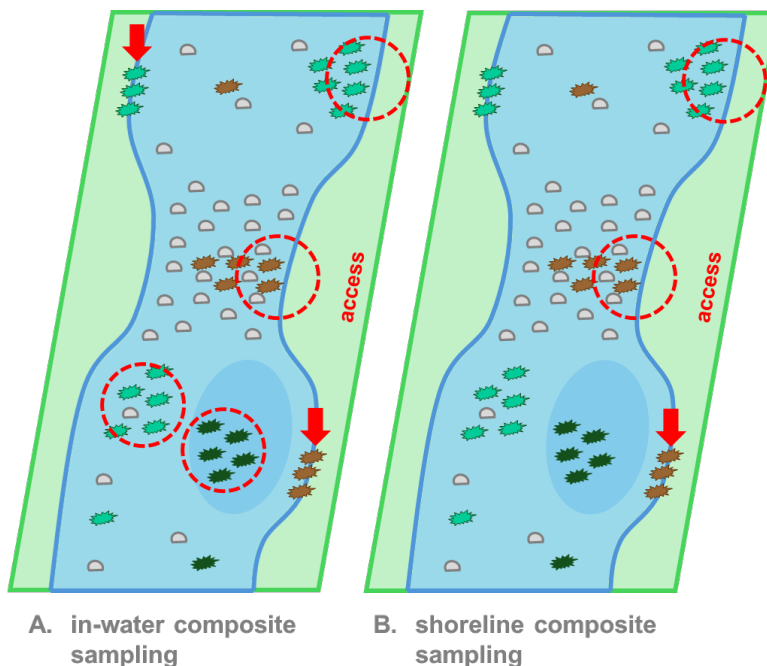


Figure 6. Schematic showing: A) in-water composite sampling of light green, brown, and dark green cyanobacterial morphotypes; and B) shoreline composite sampling of light green and brown cyanobacterial morphotypes. Cyanobacteria hotspots are shown with dashed circles and hazards are shown with red arrows. Note that more hotspots and hazards can be sampled with the in-water approach if accessing from the right side of the waterbody.

- 6.4. Record attributes regarding the sample collection beginning on **Section 5 of the Field Data Sheet** (Figure 7). Then skip step 8.



SOP - Benthic Harmful Cyanobacterial Bloom Field Procedure to Inform Water Recreation Advisories  
Version 2.0 1/22/2025

| California FHAB Program                                               |  | Benthic Cyanobacteria Field Data Sheet                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | Version 2024.09.06                                                                                                                                                         |  |
|-----------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Waterbody and site name:                                              |  | Date (mm/dd/yyyy):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | Personnel:                                                                                                                                                                 |  |
| Coordinates (dec. deg.):                                              |  | Sample Time (24:00):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | Agency:                                                                                                                                                                    |  |
| What type of field visit is being conducted? (check one)              |  | <input type="checkbox"/> <b>Event response.</b> Field visit initiated by observation or bloom report; no prior sampling or sampling frequency > 2 weeks.<br><input type="checkbox"/> <b>Follow-up monitoring.</b> Field visit initiated by previous event response; monitoring frequency is ≤ 2 weeks.<br><input type="checkbox"/> <b>Routine monitoring.</b> Field visits occur during pre-determined timeframe; monitoring occurs every 1-2 weeks.<br><input type="checkbox"/> <b>Other (please describe):</b> |  |                                                                                                                                                                            |  |
| (1) How is cyanobacteria coverage assessed in 150 ft sampling area?   |  | <input type="checkbox"/> <b>Shoreline estimate.</b> Cyanobacteria percent cover was estimated based on observations from shore.<br><input type="checkbox"/> <b>In-water estimate.</b> Percent cover was estimated by wading in zig-zag patterns within waterbody.<br><input type="checkbox"/> <b>Other (please describe):</b>                                                                                                                                                                                    |  | Percent cover diagram:                                                                                                                                                     |  |
| (2) Are cyanobacteria present?                                        |  | <input type="checkbox"/> <b>Yes.</b> Take photos of cyanobacteria. What genera are present (if known)?<br><input type="checkbox"/> <b>No.</b> Document other algae below.                                                                                                                                                                                                                                                                                                                                        |  | <input type="checkbox"/> <i>Anabaena</i><br><input type="checkbox"/> <i>Microcoleus</i><br><input type="checkbox"/> <i>Oscillatoria</i><br><input type="checkbox"/> Other: |  |
| (3) Where or how are the benthic mats growing (check all that apply)? |  | <input type="checkbox"/> in riffle <input type="checkbox"/> near shore <input type="checkbox"/> in pool or run <input type="checkbox"/> on roots, plants, or algae<br><input type="checkbox"/> on rocks <input type="checkbox"/> on sediment <input type="checkbox"/> in backwater area <input type="checkbox"/> in spires or bubble towers                                                                                                                                                                      |  |                                                                                                                                                                            |  |
| (4A) Check any of the visual hazards that apply.                      |  | <input type="checkbox"/> Cyanobacterial mats are growing in puddles on shore, or have detached and accumulated onshore.<br><input type="checkbox"/> Mats are peeling from substrates, breaking apart, or overlapping, i.e., there is risk for detachment.<br><input type="checkbox"/> Floating mats observed in stagnant areas or flowing downstream during survey.                                                                                                                                              |  |                                                                                                                                                                            |  |
| (4B) If present, how are benthic cyanobacterial mats distributed?     |  | <input type="checkbox"/> <b>Dispersed throughout reach.</b> Cyanobacteria are growing throughout the waterbody.<br><input type="checkbox"/> <b>Concentrated.</b> Cyanobacteria are localized in specific areas. How many areas?                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                            |  |
| (4C) What is size class of largest mat patch?                         |  | <input type="checkbox"/> ≤ basketball <input type="checkbox"/> ≤ doormat <input type="checkbox"/> ≤ sedan <input type="checkbox"/> > sedan <input type="checkbox"/> Other:                                                                                                                                                                                                                                                                                                                                       |  |                                                                                                                                                                            |  |
| (4D) Using diagram, what is total estimated % cover?                  |  | <input type="checkbox"/> not present <input type="checkbox"/> <15% cover <input type="checkbox"/> 15-50% cover <input type="checkbox"/> >50% cover                                                                                                                                                                                                                                                                                                                                                               |  |                                                                                                                                                                            |  |
| (5) Are benthic mats being collected for lab analysis?                |  | <input type="checkbox"/> <b>Yes.</b> Are SPATs being deployed? <input type="checkbox"/> <b>Yes.</b> Every ____ wk.<br><input type="checkbox"/> <b>No.</b>                                                                                                                                                                                                                                                                                                                                                        |  | Document SPAT location (if apply):                                                                                                                                         |  |
| (6) If collecting benthic mats, how is the sample collected?          |  | <input type="checkbox"/> <b>From the shore.</b> 2 shore hotspots x 3 pieces of mats from each hotspot (+ 3 pieces from floating mats, if present)<br><input type="checkbox"/> <b>From shore and water.</b> 2 shore hotspots + 2 water hotspots x 3 pieces each (+ 3 pieces from floating mats, if present)                                                                                                                                                                                                       |  |                                                                                                                                                                            |  |
| (7) How will samples be analyzed in the lab (check all apply)?        |  | <input type="checkbox"/> Anatoxins <input type="checkbox"/> Microcystins <input type="checkbox"/> Saxitoxins <input type="checkbox"/> Cyanobacteria ID (do not freeze)<br><input type="checkbox"/> Cylindrospermopsins <input type="checkbox"/> Nodularins <input type="checkbox"/> qPCR (do not freeze) <input type="checkbox"/> Other:                                                                                                                                                                         |  |                                                                                                                                                                            |  |
| Other field data (if collected)                                       |  | Water temp (°C): Sp. cond (µS/cm): Salinity (ppt): Diss. O2 (%): Diss. O2 (mg/L): pH: pH (mV): Turbid (NTU): Bar. pressure:                                                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                                            |  |
| Additional notes or observations:                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                            |  |

Figure 7. Area in the field data sheet (shown in red square) for recording information on the type of samples being collected, how benthic mats are being sampled (if any), and the type of lab analyses being ordered (if known).

7. Option C. Shoreline procedure for Mat Collection (For in-water procedure, go to step 6)
  - 7.1. **Collect the composite sample at the shoreline.** Fill the sample ¼ of total volume with ambient water to provide moisture to collected material. Walk back along the surveyed area and collect the sample from 2-4 hotspots (Figure 6B). Hotspots are locations that you observe to have the densest or most area of coverage of cyanobacteria growth.
  - 7.2. At each hotspot, collect 3 pieces of mat material (each approx. 3 cm<sup>2</sup>) by pinching the material with gloved hands and dropping into the same sample container. Target mats with different appearances to produce a composite sample of diverse genera. In sum, 12 pieces of mat material are composited into the same sample container. If there are fewer than 4 hot spots in the sampling area, collect additional pieces at the identified hot spots to equal a total of 12 pieces collected for the composite sample.
    - 7.2.1. Collection of a diverse population of mat materials is the goal of this composite sample. Target sample collection of visually distinct mats, to avoid collecting pieces of the same type of mat. Many species of cyanobacteria have different shades of color, texture, and density. If mat material appears visually uniform, collect pinches of material that are not side-by-side or connected to each other.
    - 7.2.2. Also, target mat material should include visual hazards like floating mats or those accumulated along the shoreline, as well as mats that are attached to the bottom substrate (cobble, sand).
  - 7.3. If mat material cannot be pinched to remove it, use a scraper, forceps, turkey baster (when material is submerged), or other tool to grab approx. 3 cm<sup>2</sup> (approx. size of a quarter) of material to replicate a “pinch.” If collection of mat materials is



challenging, the sampler can use a bucket or bin to collect larger pieces of mat materials and then take the bin to a staging area to retrieve 3 pinched pieces of mat material for the sample container. Avoid collecting detritus, sand, pebbles, or invertebrates with your sample.

- 7.4. Record attributes about the sample collected beginning on **Section 5 of the Field Data Sheet** (Figure 7).
- 7.5. If the mat material is inaccessible (not near the shoreline or within distance to grasp material without entering the water), then record visual observations to the field data sheet and note that no sample was collected. Then go to the [reporting step](#).
8. Labeling and preservation of sample
  - 8.1. Close sample container. Verify that all material is submerged in ambient water. If the material is not submerged, use forceps or a stick to submerge material. If the sample does not contain enough water to submerge mat material, add ambient water sparingly.
  - 8.2. If the label is not pre-attached, wipe off the exterior of the sample container and add the label.
  - 8.3. Place a sample container into a cooled ice chest. The sample containers should be stored in the dark and be cooled to 2-8°C (do not freeze) during the remainder of the field sampling event.
  - 8.4. Fill out the COC form and ensure all information on the labels is correct and consistent with the information on the COC form and the field sheet. Take a photo of the COC form to submit to the lab as notification of shipping samples and place the COC form in a resealable zipper plastic bag.
  - 8.5. Other field parameters as well as any notes and observations can be recorded in the final sections of the Field Data Sheet. Notes and observations may include presence of non-toxic algae or if cyanobacteria are inaccessible (Figure 8).

| California FHAB Program                                               |  | Benthic Cyanobacteria Field Data Sheet                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | Version 2024.09.06                                                                                                                                    |  |
|-----------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Waterbody and site name:                                              |  | Date (mm/dd/yyyy):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | Personnel:                                                                                                                                            |  |
| Coordinates (dec. deg.):                                              |  | Sample Time (24hrs):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | Agency:                                                                                                                                               |  |
| What type of field visit is being conducted? (check one)              |  | <input type="checkbox"/> <b>Event response.</b> Field visit initiated by observation or bloom report; no prior sampling or sampling frequency > 2 weeks.<br><input type="checkbox"/> <b>Follow-up monitoring.</b> Field visit initiated by previous event response; monitoring frequency is ≤ 2 weeks.<br><input type="checkbox"/> <b>Routine monitoring.</b> Field visits occur during pre-determined timeframe; monitoring occurs every 1-2 weeks.<br><input type="checkbox"/> <b>Other</b> (please describe): |  |                                                                                                                                                       |  |
| (1) How is cyanobacteria coverage assessed in 150 ft sampling area?   |  | <input type="checkbox"/> <b>Shoreline estimate.</b> Cyanobacteria percent cover was estimated based on observations from shore.<br><input type="checkbox"/> <b>In-water estimate.</b> Percent cover was estimated by wading in zig-zag patterns within waterbody.<br><input type="checkbox"/> <b>Other</b> (please describe):                                                                                                                                                                                    |  | Percent cover diagram:                                                                                                                                |  |
| (2) Are cyanobacteria present?                                        |  | <input type="checkbox"/> <b>Yes.</b> Take photos of cyanobacteria. What genera are present (if known)?<br><input type="checkbox"/> <b>No.</b> Document other algae below.                                                                                                                                                                                                                                                                                                                                        |  | <input type="checkbox"/> Anabaena<br><input type="checkbox"/> Microcoleus<br><input type="checkbox"/> Oscillatoria<br><input type="checkbox"/> Other: |  |
| (3) Where or how are the benthic mats growing (check all that apply)? |  | <input type="checkbox"/> in riffle<br><input type="checkbox"/> on rocks<br><input type="checkbox"/> near shore<br><input type="checkbox"/> on sediment<br><input type="checkbox"/> in pool or run<br><input type="checkbox"/> in backwater area<br><input type="checkbox"/> on roots, plants, or algae<br><input type="checkbox"/> in spires or bubble towers                                                                                                                                                    |  |                                                                                                                                                       |  |
| (4A) Check any of the visual hazards that apply.                      |  | <input type="checkbox"/> Cyanobacterial mats are growing in puddles on shore, or have detached and accumulated onshore.<br><input type="checkbox"/> Mats are peeling from substrates, breaking apart, or overlapping, i.e., there is risk for detachment.<br><input type="checkbox"/> Floating mats observed in stagnant areas or flowing downstream during survey.                                                                                                                                              |  |                                                                                                                                                       |  |
| (4B) If present, how are benthic cyanobacterial mats distributed?     |  | <input type="checkbox"/> <b>Dispersed throughout reach.</b> Cyanobacteria are growing throughout the waterbody.<br><input type="checkbox"/> <b>Concentrated.</b> Cyanobacteria are localized in specific areas. How many areas?<br>In addition to above, check here if waterbody is too wide (>30 ft) for shoreline visuals or wading                                                                                                                                                                            |  |                                                                                                                                                       |  |
| (4C) What is size class of largest mat patch?                         |  | <input type="checkbox"/> ≤ basketball<br><input type="checkbox"/> ≤ doormat<br><input type="checkbox"/> ≤ sedan<br><input type="checkbox"/> > sedan<br><input type="checkbox"/> Other:                                                                                                                                                                                                                                                                                                                           |  |                                                                                                                                                       |  |
| (4D) Using diagram, what is total estimated % cover?                  |  | <input type="checkbox"/> not present<br><input type="checkbox"/> <15% cover<br><input type="checkbox"/> 15-50% cover<br><input type="checkbox"/> >50% cover                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                       |  |
| (5) Are benthic mats being collected for lab analysis?                |  | <input type="checkbox"/> <b>Yes.</b><br><input type="checkbox"/> <b>No.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      |  | <input type="checkbox"/> <b>Are SPATs being deployed?</b><br><input type="checkbox"/> <b>Yes.</b> Every wk.<br><input type="checkbox"/> <b>No.</b>    |  |
| (6) If collecting benthic mats, how is the sample collected?          |  | <input type="checkbox"/> <b>From the shore.</b> 2 shore hotspots x 3 pieces of mats from each hotspot (+ 3 pieces from floating mats, if present)<br><input type="checkbox"/> <b>From shore and water.</b> 2 shore hotspots + 2 water hotspots x 3 pieces each (+ 3 pieces from floating mats, if present)                                                                                                                                                                                                       |  |                                                                                                                                                       |  |
| (7) How will samples be analyzed in the lab (check all apply)?        |  | <input type="checkbox"/> Anatoxins<br><input type="checkbox"/> Cyanobacteriosporins<br><input type="checkbox"/> Microcystins<br><input type="checkbox"/> Nodularins<br><input type="checkbox"/> Saxitoxins<br><input type="checkbox"/> qPCR (do not freeze)<br><input type="checkbox"/> Cyanobacteria ID (do not freeze)<br><input type="checkbox"/> Other:                                                                                                                                                      |  |                                                                                                                                                       |  |
| Other field data (if collected)                                       |  | Water temp (°C):   Sp. cond (µS/cm):   Salinity (ppt):   Diss. O2 (‰):   Diss. O2 (mg/L):   pH:   pH (mV):   Turbid (NTU):   Bar. pressure:                                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                       |  |
| Additional notes or observations:                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                       |  |

Figure 8. Area of the field data sheet (shown in red square) for recording other field parameters (if collected) as well as any notes and observations.

## Sample Handling and Shipping

*This section provides the steps for handling, storing, and shipping samples.*

9. Sample handling and shipping.
  - 9.1. Sample preservation and containers. Samples should be stored in controlled environments during field visits and when being shipped to the laboratory. Benthic HCB samples are sensitive to high temperatures and bright light. Store samples under cool temperatures 2-8° C (exceeding 10° C can degrade samples) in dark or amber-colored *glass* containers. Plastic containers can adsorb toxins produced by cyanobacteria causing low quality data.
  - 9.2. Sample preservation. The samples should be cooled to 2-8°C as soon as possible after sampling. On hot days, replenish gel packs in the ice chest prior to shipping to ensure that samples remain cool.
  - 9.3. Sampling holding time. The time from sample collection to lab analysis start time should not exceed 5 days. Contact the lab to learn how much time they need from sample receipt to sample analysis start. Best practices include shipping samples the same day they are collected or the next day to the lab.
    - 9.3.1. If shipping the same day, pack the shipping container with fresh/cold ice packs that were not used to cool the samples during the day. Adequate gel packs should be used to maintain cool temperatures (<10°C).
    - 9.3.2. If samples are not shipped the same day, then store samples in the dark and refrigerate and ship the samples the following day.
    - 9.3.3. If samples are frozen, several analyses by the laboratory cannot be performed (e.g., qPCR, taxonomy) so plan according to project goals. Frozen samples can be stored at - 80°C for up to 6 months. Glass bottles should not be frozen unless 1/2 of its volume is empty to allow for liquid-to-solid expansion during freezing.
  - 9.4. Sample shipping. *Ship samples overnight* with a courier to the testing lab to maintain temperature. Place the chain of custody (COC) form within a plastic bag (to prevent damage) and include within the shipping container. Make a copy of the COC form for your own records (a photo from a smartphone is adequate). If the shipping container must be returned, include a prepaid shipping label or arrange for the laboratory to return the container for an additional fee
    - 9.4.1. Avoid shipping with wet ice, loose or bagged, due to the possibility of contamination and leaks during shipment. Glass sampling bottles should be packed with extra cushioning to prevent movement of bottles and breakage (ice packs are not sufficient cushioning material, use bubble wrap). During warmer months, plan for additional ice packs to ensure samples stay below temperature.
  - 9.5. Sample drop off. If samples will be transported by personnel and dropped off the same day to the laboratory, check for their latest time for drop off to avoid closures.

## Equipment Cleaning

10. Cleaning of equipment to prevent cross contamination and spread of invasive species.
  - 10.1. When sampling instruments will not be reused on the same day. If the sampling instrument is not reused during a field trip, rinse its components thoroughly with clean water before they dry and place the sampler in a plastic bag or wash bin for transport to the office/laboratory for cleaning.
  - 10.2. If sampling instruments will be reused. If the sampling instrument is reused during the field trip, rinse its components with clean water before they dry. Place all components of sampling equipment into a plastic bag or wash bin for transport to the next sampling site.
  - 10.3. Field-clean all the sampling equipment at the next sampling site using 0.2 % Liquinox™ detergent (or other phosphate free detergent); soak and scrub parts in wash basin; rinse with clean water; then rinse with native water before use (EPA, 2015; Graham, 2008). If using a hand-held sonde or other equipment that is being used across multiple waterbodies, use best practices to prevent transmission of invasive species and cross-contamination.

## Laboratory Communication and Analysis

*Guidance for toxin analysis:* Request the laboratory to measure the total fraction of toxins when conducting water monitoring to inform recreational health advisories. The total fraction includes free toxins (extracellular) dissolved in water and any toxins contained within cyanobacteria cells and mats (intracellular). Among cyanobacteria producers of microcystins, nodularin, anatoxin-a, and saxitoxin, a greater proportion of toxins are contained within cells and mats.

*Guidance for laboratories conducting toxin analysis:* In advance of sampling, provide the laboratory with a copy of the SOP for Lab Processing of Benthic HCBs (Benthic Lab SOP) developed by the Water Boards Surface Water Ambient Monitoring Program found at: [My Water Quality: California Harmful Algal Blooms \(HABs\)](#). The Benthic Lab SOP standardizes processing of mats prior to toxin or DNA analysis to better mix the heterogeneous sample and report toxin analysis results in appropriate units by dry weight of mass (toxin/dry weight of sample) to inform decision making. The Benthic Lab SOP also includes an optional add-on procedure for processing benthic samples for DNA analysis.

## Reporting

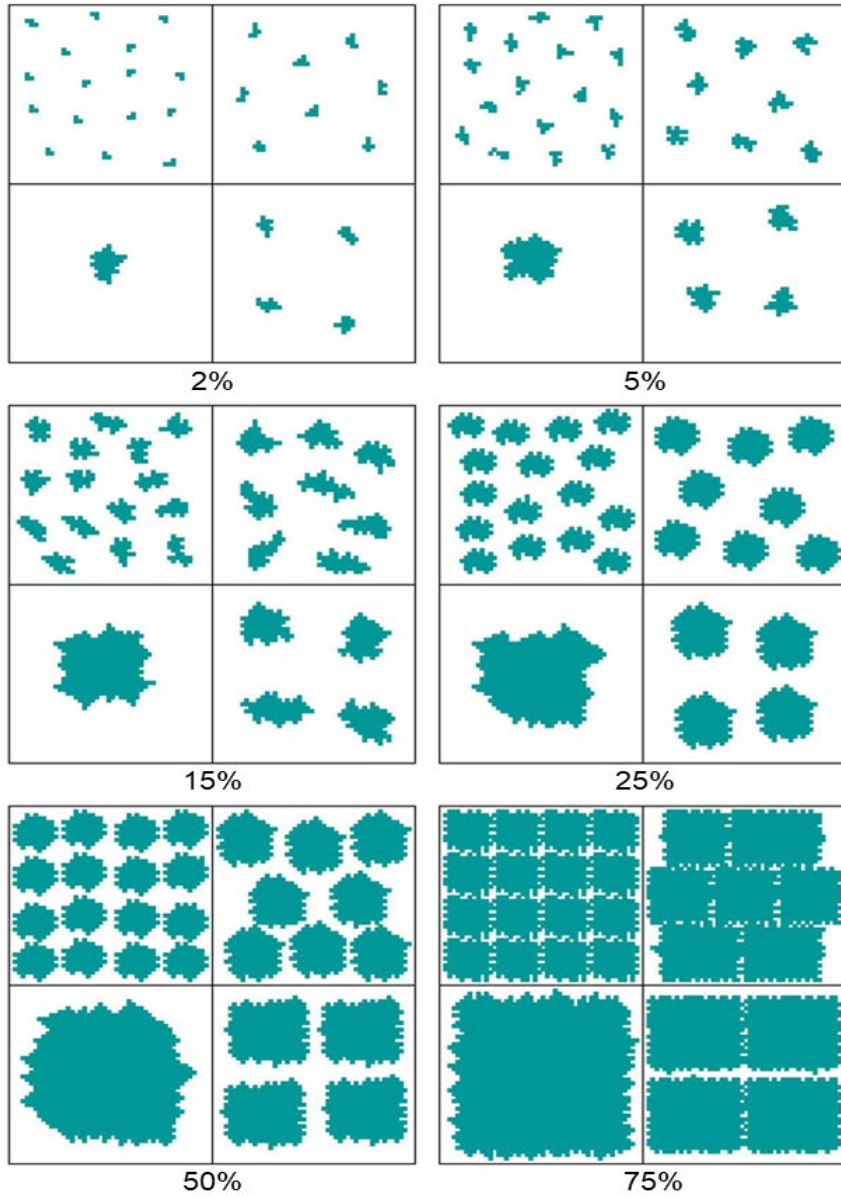
CCHAB Network requests users of this SOP to share their monitoring results to the Water Boards HAB Program at [cynoHAB.reports@waterboards.ca.gov](mailto:cynoHAB.reports@waterboards.ca.gov). The HAB Program implements CCHAB Network's guidance statewide and will provide training to users on implementing this SOP and interpreting the results. Any shared data will be published to a public web map after receiving permission from the user.

In addition, the state data repository for water quality and HCB monitoring is housed in CEDEN database ([CEDEN - California Environmental Data Exchange Network](#)). The CCHAB Network

relies on monitoring entities to submit data to a centralized repository to inform statewide assessments and reference when requesting funding to address HCBs.

## Attachment A

**Figure 9. Diagram to reference for estimating percent cover.**



## Attachment B

This is an excerpt from the standard operating procedure (SOP) for collecting benthic (bottom) HCB measurements and samples in the field and developed as part of the [Benthic HCB Guidance](#). Monitoring HCBs using standardized processes is critical to evaluating water quality conditions that inform the potential exposure and health impacts associated with water recreation and inform decisions about posting advisories.

### Field Procedures Overview

This procedure can be conducted on a flowing channel (e.g., streams, rivers) or along the beach of a ponded waterbody (e.g., lake, reservoir, pond). Key concepts are listed as follows:

- Sampling area – Size of sampling area is approximately 150 linear ft. of beach or channel. Recreational areas should be targeted but if the substrate was previously disturbed, move the sampling area upstream where potential benthic HCB growth is observed.
- Data collection and measurements - Personnel will record observations and percent coverage of mats in the field data sheet ([Appendix B](#)). Visual hazards (e.g., detached and floating mats) associated with benthic HCBs will also be recorded in the field data sheet.
- Number of samples collected - Personnel will collect 1 composite mat sample.

Monitoring entities have 3 procedure options to choose from based on waterbody types and resource constraints. Options A and B are combinations of in-water and shoreline evaluations. Due to constraints of monitoring entities in California, Option C does not require field personnel to enter the water.

**Table 1. Field Procedure Options**

| <b>Field Procedure (pick one)</b>                                              | <b>Percent Coverage</b><br>(categories: 0%/not present, <15%, 15-50% and greater than 50%)                                                                                                                                                                            | <b>Composite mat sample</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. Combination of <i>In-water and shoreline</i> procedure for flowing channels | Percentage cover is visually estimated in areas with the most spatial coverage of mats observed along the shoreline and observed in-water. Hot spots are identified after surveying the full sample area. (Figure 3A, 3B)                                             | Fill container ¼ full with ambient water. Collect mat samples at 4 hot spots, 2 hot spots observed along the shoreline and 2 hot spots observed in-water. Hot spots can include detached and floating mats found in-water or on the shoreline. At each hot spot collect 3 pieces (approx. 3 cm <sup>2</sup> ) of visually distinct mats to collect diverse material. (Figure 6A). Total of 12 pieces collected for the composite.                                      |
| B. Combination of <i>In-water and shoreline</i> procedure for ponded waterbody | Percentage cover is visually estimated in areas with the most spatial coverage of mats observed along the shoreline and observed in-water. Hot spots are identified after surveying the full sample area. (Figure 3A, 3B) Limit depth of water to knee deep or 0.5 m. | Fill container ¼ full with ambient water. Collect mat samples at 4 hot spots, 2 hot spots observed along the shoreline and 2 hot spots observed in-water. Hot spots can include detached and floating mats found in-water or on the shoreline. At each hot spot collect 3 pieces (approx. 3 cm <sup>2</sup> ) of visually distinct mats to collect diverse material. (Figure 6A). Total of 12 pieces collected for the composite.                                      |
| C. <i>Shoreline only</i> procedure for both waterbody types <sup>16</sup>      | Percentage cover is visually estimated at areas with the most lateral growth observed along the shoreline and near shore area. Hot spots are identified after surveying the full sample area. (Figure 3C)                                                             | Fill container ¼ full with ambient water. Collect mat samples at 2-4 hot spots along the shoreline. Hot spots can include detached and floating mats found in near shore area or on the shoreline. At each hot spot collect 3 pieces (approx. 3 cm <sup>2</sup> ) of visually distinct mats to collect diverse material. (Figure 6B). If there are fewer than 4 hot spots, collect additional pieces at hot spots to a total of 12 pieces collected for the composite. |

<sup>16</sup> Due to constraints of monitoring entities in California, this provides an option for field personnel to not enter the water due to lack of authorization or field supplies. If this is not a constraint for your monitoring entity, implement in-water Options A or B.



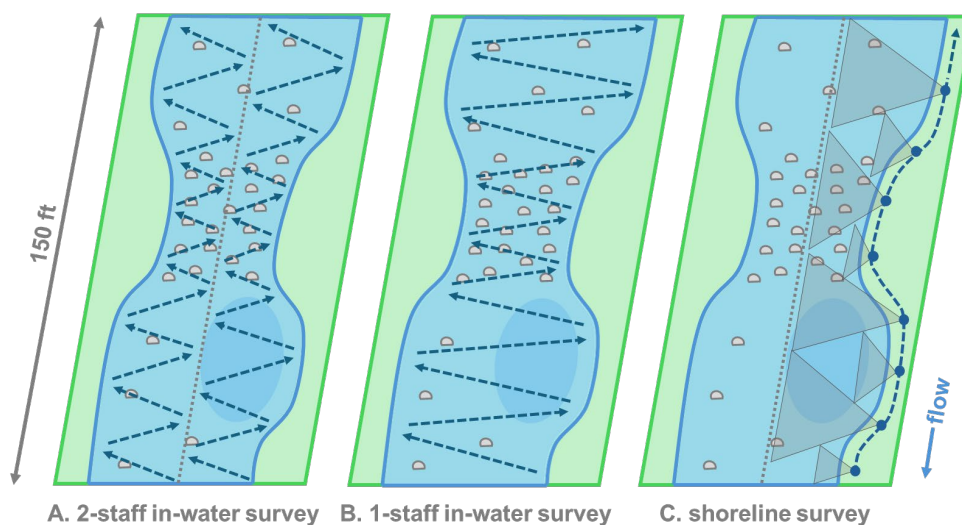


Figure 3. Schematic for conducting: A) in-water survey with 2-personnel wading in zig-zag patterns; B) in-water survey with 1-personnel wading in a zig-zag pattern; and C) shoreline survey with periodic stops for assessing nearshore and at least 30 ft offshore. .

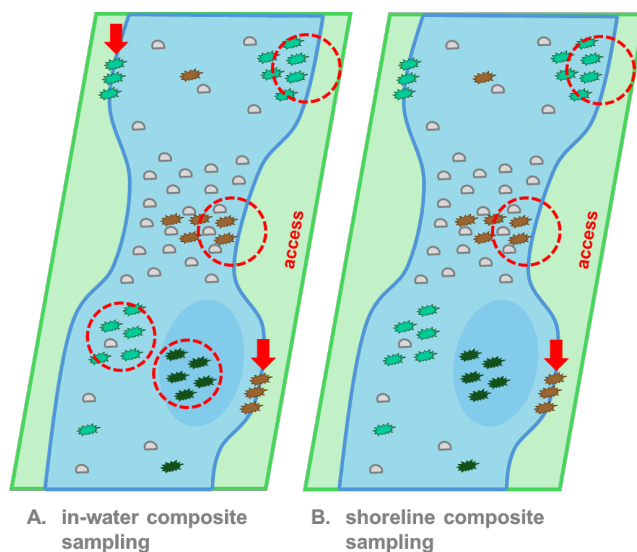


Figure 6. Schematic showing: A) in-water composite sampling of light green, brown, and dark green cyanobacterial morphotypes; and B) shoreline composite sampling of light green and brown cyanobacterial morphotypes. . Cyanobacteria hotspots are shown with dashed circles and hazards are shown with red arrows. Note that more hot spots and hazards can be sampled with the in-water approach if accessing from the right side of the waterbody.